Keeping the sign of the threshold in the fixed-target resource

Bartosz Kołodziejek and Kacper Rodziewicz

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This research note proves a conditional improvement to the coarse-entry estimate. Its purpose is to retain information already paid for in that computation. It does not prove the remaining confinement, count comparison, or finite base. AI assisted its derivation, implementation and exposition; the accompanying independent checks have the specific scope described below.

1 Why the threshold matters

The coarse entry used $|H|$ to obtain one bound valid for both signs of the raw sum threshold. This was convenient for acquiring moments, but loses information needed at the next step. A term KH can help when $H > 0$ and hurt when $H < 0$. Replacing it by $|K|\,|H|$ too early discards that help. We retain its sign within each previously accepted moment cell.

Fix $n \geq N = 42$ and $c = C_{\mathrm{E}}$. Let P be an attained fixed-target maximizing witness with nonnegative excess

$$a = \sqrt{n}\,\{\mathbb{P}(S_n \leq H) - \Phi(H/\sqrt{n})\} - c\rho \geq 0, \qquad \rho = \mathbb{E}|X|^3.$$

Retain the predecessor, contact, gap, and Fourier hypotheses of `INITIAL42_FIXED_TARGET_COARSE_V7`. Write $\gamma = \mathbb{E}X^3$, $\beta = \mathbb{E}[X|X|]$ and

$$D = (3+\sqrt{10})\rho - \gamma - 3J, \qquad \mathcal{U} = (n+H^2)a + n\phi(0)D/6 + n^{3/2}L \geq 0.$$

Here J is the actual weighted gap sum and L is the nonnegative leakage from the conservation identity. Neither may be replaced by the corresponding quantity of a two-point approximation.

2 A signed bound from one accepted cell

The fourth signed pass of the entry proof already gives, on each cell,

$$\mathcal{R}_c \leq K_F + cA + (cK_1 + \varphi W)H + s|H| - (c\rho + \varphi\gamma/2)H^2, \qquad \varphi = \phi(H/\sqrt{n}). \tag{1}$$

The intervals for K_1, W, ρ, γ and φ refer to the same law. The letter W here denotes the signed moment expression in the remainder; it is not a conditional noise variable. The cell supplies $\rho \geq r_l$, $\gamma \geq g_l$, $\varphi \in [\varphi_l, 2/5]$, and a previously proved raw ceiling $\mathcal{R}_c \leq T < Nc_-r_l$, where $c_- \leq c \leq c_+$.

There are two ordered uses of the raw estimate. The density floor in (1) uses the preceding pass. The newly obtained ceiling T is then used in conservation to pay for an additional quadratic term. In detail, put $w = H^2/(2n)$ and $M = c\rho - \phi(0)D/6$. The conserved inequality gives $e^{-w}M \geq c\rho - T/n > 0$ and

$$\mathcal{R}_c - \mathcal{U} = n(1-e^{-w})M \geq \frac{H^2}{2}(c\rho - T/n) \geq \eta H^2, \qquad \eta = \frac{c_-r_l - T/N}{2} > 0.$$

We used $e^w - 1 \geq w$. This argument does not use its conclusion to justify its own density floor.

For $t = c_-$ or c_+, define

$$B_t = K_F + tA, \quad Q_t = tr_l + \min_{\varphi\in[\varphi_l,2/5]} \varphi g_l/2 + \eta,$$

$$F_{t,+} = \max(tK_1 + \varphi W) + s, \qquad F_{t,-} = -\min(tK_1 + \varphi W) + s.$$

The maxima and minima range over the recorded rectangular intervals. They can be evaluated at the eight corners, since the expression is affine in each variable separately. All exported Q_t are positive.

Proposition 1 (Signed resource)**.** *For $h = |H|$ and the original sign ϵ of H,*

$$0 \le \mathcal{U} \le \min\left\{T, \max_{t\in\{c_-,c_+\}} (B_t + F_{t,\epsilon}h - Q_t h^2)\right\}.$$

At $H = 0$ either sign gives the same bound.

Proof. For either fixed sign, the maximum defining $F_{t,\epsilon}$ is a maximum of affine functions of t, hence is convex in t. For fixed $h \ge 0$, $B_t + F_{t,\epsilon}h - Q_t h^2$ is therefore convex. Its maximum on the enclosing interval for c occurs at an endpoint. Combining (1) with the ordered phase price proves the quadratic bound. Also $\mathcal{U} \le \mathcal{R}_c \le T$. A negative signed slope remains negative; replacing it by zero would lose a valid improvement. □

3 Using a noise restriction without fixing the index

Let $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $v = pq$, and $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$. The possible atom at zero is assigned to the nonpositive class. Conditional centering gives

$$\theta = 1 - \frac{m^2}{v}, \qquad \ell^2 = \frac{1-\theta}{v}, \qquad \lambda = \frac{n\theta}{\ell^2} = \frac{nv\theta}{1-\theta}.$$

Consequently

$$\theta = \frac{\lambda}{nv + \lambda}, \qquad m^2 = \frac{v^2}{v + \lambda/n}. \tag{2}$$

The two unnormalized conditional variance contributions θ_+, θ_- are nonnegative and sum to θ. Thus

$$\beta = (1-\theta)(1-2p) + \theta_+ - \theta_-, \qquad 1 - 2p - 2q\theta \le \beta \le 1 - 2p + 2p\theta.$$

These formulas retain the orientation when $p > 1/2$.

Lemma 1 (A necessary filter)**.** *Let $p \in [l,u] \subset (0,1)$ and $0 \le \lambda \le \Lambda$. Put $v_- = \min\{l(1-l), u(1-u)\}$ and let v_+ be the maximum of these two values and $1/4$ if $[l,u]$ contains $1/2$. For every $n \ge 42$,*

$$\theta \le \Theta = \frac{\Lambda}{42v_- + \Lambda}, \qquad \sqrt{\frac{v_-^2}{v_- + \Lambda/42}} \le m \le \sqrt{v_+},$$

$$1 - 2u - 2(1-l)\Theta \le \beta \le 1 - 2l + 2u\Theta.$$

Proof. The function $v^2/(v+s)$ increases with $v > 0$ and decreases with $s \ge 0$. Apply this fact to (2), then enlarge the coefficients in the two beta inequalities outward on $[l,u]$. This includes the limit $n \to \infty$; it is not a fixed-index approximation. □

For a closed region $p \in [l,u]$, $h \in [h_0, h_1]$, $\lambda \in [0,\Lambda]$, retain every cell meeting these necessary ranges and with its acquired phase ceiling at least h_0. Closed overlaps are retained. For each retained cell clip h_1 by that phase ceiling. A concave quadratic $B + Fh - Qh^2$, $Q > 0$, has its maximum at

$$h = \max\{h_0, \min(h_1, F/(2Q))\}.$$

Apply this to both target endpoints, intersect with that cell's T, and take the maximum over all retained cells. An empty cell list excludes the region only when the complete ancestor cover has first been established. A negative resulting upper bound contradicts $\mathcal{U} \ge 0$.

4 Exact closed-region consequences

The complete fourth pass has 2,048 boxes and exports all 682 active cells. The following are outward rational upper bounds with denominator 10^9; all rows have $p \in [.41, .42]$. No sampled phase value is substituted for an interval.

$\lvert H\rvert$	Λ	cells	$H \geq 0$	$H \leq 0$
1.25	unrestricted	331	1.032820261	1.266287022
1.25	.15	15	−.128026665	.206210125
$[1.24, 1.26]$	.15	18	−.094649077	.218329547
$[1.20, 1.30]$	.15	22	−.033287706	.265382458

In particular, under the retained entry hypotheses, no nonnegative-excess witness can have

$$p \in [.41, .42], \qquad H \in [1.20, 1.30], \qquad 0 \leq \lambda \leq .15.$$

For the reflected sign the last row is a resource ceiling, not an exclusion. A bound on one such region does not prove full confinement.

5 What is accepted, and what remains

The export adds three reversible insertions to the pinned signed kernel: initialization, an append after each surviving cell, and a return field. It changes none of the old bounds or rejection branches. Its exact reverse recovers the original source. The producer replays the final pass from the accepted third-pass rows, checks every resulting row against the selected entry object, and records all active cells. It does not claim to repeat the accepted prefix.

The separate acceptance program pins its sources and inherited inputs before executing them, replays all 2,048 boxes, and compares the complete ordered JSON object with the proposed export. Comparison uses serialized types, so an integer cannot be replaced by a Boolean or a floating value. Duplicate keys, missing, repeated or reordered cells, false bounds and extra fields are rejected. This proves agreement with the reviewed deterministic computation, not the correctness of its analytic premises.

The independent mathematical review reconstructs all 1,364 endpoint quadratics by a different rectangular-support formula, checks every phase price and preceding density floor, and evaluates all nine recorded closed regions by endpoints and interior vertices. It imports no inherited kernel. A separate executable review addresses source binding and coverage. The original selected entry replay, contact/conservation arguments, support acquisition, and complete Fourier estimates remain dependencies.

Keeping these signed envelopes requires no new Fourier integration beyond the last accepted pass. It removes the need to replace that pass by a single worst-case resource in these downstream estimates. It does not remove a complete confinement job, count theorem, finite certificate, or any existing repository proof object.